FACTOR

Dominican digital ID

Action points and questions for the JCE:
certificate, trust and signer

FACTOR INC. · CRYPTOGRAPHIC AUDIT

projectfactor.io

Dominican digital ID

Action points and questions for the JCE: certificate, trust and signer

Prepared from the technical analysis of a production cédula. It contains no personal data of the holder; only public technical identifiers are used.

Requested decision. The production card is *fully personalized*: it contains two private keys of the holder (signature and authentication, non-exportable) and all PINs/PUKs are set; its identity integrity was verified. Three points remain for the JCE: (1) the holder's *certificate* is not on the card; (2) the issuer's trust cannot be independently validated; and (3) a development signer persists in production.

Findings (summary, no personal data)

Status	Finding	Evidence basis
Verified	The integrity of the identity data is verified: the CMS signature of the EF.SOD is valid and the 7 readable data groups (DG1, 2, 7, 11–14) reconcile with the SOD under SHA-256.	EF.SOD signer: `JCE DS 01`, ECDSA/SHA-384.
Verified	The card is **personalized**: two private keys of the holder —signature (ref. `0x41`, signature PIN `0x22`) and authentication (ref. `0x42`, user PIN `0x05`)—, both **non-exportable**; all PINs/PUKs are set.	PrKDF: 2 keys; AODF: 6 objects; PIN status (ref. 83/84/92/93) = set with full retry counters; signature DF `0DF5` present.
Condition	There is no **X.509 certificate** of the holder on the card (the CDF directory is empty and no certificate EF was found). This is observed equally on the other cards examined, so the certificate is probably managed off the card or its issuance is pending.	CDF (4404) = 250 B of zeros; no certificate EF under `0DF5`; PuKDF empty.
Condition	**PIN provenance.** The signature PIN and the user PIN are set, but it cannot be determined whether they correspond to values chosen by the holder or to a personalization default.	Status query `63Cx` (set, without consuming retries); the value is not readable.
Condition	**The issuer's trust cannot be validated:** the cédula CSCA distribution point indicated in the certificate does not resolve in public DNS.	DS AKI `e8090eb7...`; URI `https://ca.jce.gob.do/csca` returns NXDOMAIN (the same host also serves the DS CRL).
Defect	Active **RSA-1024 development Document Signer** in production within EF.CardSecurity (obsolete algorithm and "development" provenance), identical on the cards sampled.	Subject `OU=development, O=cryptovision`; RSA 1024; cert. fingerprint `4eb279...`.

Status	Finding	Evidence basis
Action	Access to the biometrics (DG3) is correctly closed after terminal authentication (EAC); the card requires a certificate chain up to its declared anchor.	`SELECT DG3` returns `6982`; EF.CVCA declares CAR `DOCVCA00001`.

Legend: Confirmed defect | Condition / to be resolved | Context / action | Verification passed

Action points and questions

1. **Holder's certificate.** The card's certificate directory (CDF) is empty on the cards analyzed, even though the signature key and its PIN exist. Where is the holder's X.509 certificate issued and kept —is it loaded onto the card, managed off the card, or is its issuance pending— and how does a verifying party obtain it?

2. **PIN provenance and holder verification (security question).** The signature PIN and the user PIN are already set. Are they set to values chosen by the holder through an in-person, strong identity verification, or to a personalization default? In the latter case, what prevents a third party in possession of the card from using those credentials?

3. **Publication of the cédula CSCA.** So that any verifying party can validate passive authentication, we request that the cédula CSCA (SKI `e8090eb7...`) and its CRL be published or provided through a resolvable channel, given that the URI embedded in the DS certificates currently does not resolve.

4. **Remediation of the document signer.** We recommend replacing the RSA-1024 development Document Signer (`OU=development`) present in EF.CardSecurity with a production signer anchored in the JCE with strength ≥ EC-256 / RSA-3072, and removing the development CSCA from every trust store.

Additional point (optional): for biometric verification (DG3) by authorized inspection systems, the JCE would have to provision a chain of card-verifiable certificates (CVC) from the DOCVCA00001 authority. No cryptographic material or keys are requested; only procedural guidance.